KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES
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Programme : B.Tech **Academic Year**: 2025-26
Year / Semester **:** II / II
Branch / Section : IT/CSE
Course Code : 23CD4T02
Subject : Database Management System

QUESTION BANK FOR MID-II

UNIT -3
Short answer questions

Q. No.	Question	M	CO	LL
1	Differentiate between Subquery and Correlated Subquery	**2**	3	LL1
2	List aggregate functions in SQL.	**2**	3	LL1
3	Differentiate between WHERE and HAVING clause.	**2**	3	LL1
4	Define joins in SQL and list different types of joins.	**2**	3	LL1
5	Define a View in SQL. Give the syntax for creating a view.	**2**	3	LL1

Essay questions

Q. No.	Question	M	CO	LL
1	Define a nested query and sub query. Explain the use of IN, ANY, ALL, and EXISTS operators in nested queries with suitable examples.	**6**	3	LL2
2	Explain Aggregate Functions. Discuss GROUP BY and HAVING clauses in SQL with suitable examples.	**6**	3	LL2
3	Explain different types of SQL joins with syntax and examples.	**6**	3	LL2
4	Define a view in SQL and explain its operations with examples.	**6**	3	LL2
5	Explain relational set operators in SQL with examples	**6**	3	LL2
6	a) Write a SQL query with a complex WHERE clause to retrieve employees whose salary is between 50000 and 70000 (inclusive), whose last name does not start with ‘S’, and whose first name contains ‘a’ from EMPLOYEE TABLE b) Write a SQL query to find the 2nd highest salary from EMPLOYEE table using a subquery.	**6**	3	LL3

UNIT-4
Short answer questions

Q. No.	Question	M	CO	LL
1	Define Normalization and state its advantages	**2**	4	LL1
2	Define Super Key and Candidate Key.	**2**	4	LL1
3	State Functional Dependency and list its properties.	**2**	4	LL1
4	Differentiate between Trivial and Non-Trivial Functional Dependency	**2**	4	LL1
5	Define Fully Functional Dependency	**2**	4	LL1
6	Define Partial Functional Dependency	**2**	4	LL1

7	Define Transitive Functional Dependency	**2**	4	LL1
8	Define First Normal Form	**2**	4	LL1
9	Define Multivalued Dependency.	**2**	4	LL1
10	Define Lossless Join and Dependency Preserving Decomposition	**2**	4	LL1

Essay questions

Q. No.		Question	M	CO	LL
1	**a**	Define Normalization. Explain the purpose of normalization with suitable examples.	**6**	4	LL2
	b	Explain various keys in DBMS with examples.	**6**	4	LL2
2	**a**	Define Functional Dependency. Explain the different types of Functional Dependencies with examples.	**6**	4	LL2
	b	Discuss the Properties of Functional Dependency.	**6**	4	LL2
3	**a**	Explain Second Normal Form (2NF) with a suitable example.	**6**	4	LL2
	b	Explain Third Normal Form (3NF) with a suitable example.	**6**	4	LL2
4	**a**	Explain Boyce–Codd Normal Form (BCNF) with a suitable example.	**6**	4	LL2
	b	Explain the lossless and dependency preserving decomposition.	**6**	4	LL2
5	**a**	Discuss Fourth Normal Form (4NF) with a suitable example.	**6**	4	LL2
	b	Discuss Fifth Normal Form (5NF) with a suitable example.	**6**	4	LL2
6	**a**	Given relation R(A,B,C,D) has the FD {A->B, C->D}. Is this relation in BCNF? Justify your answer with a proper explanation of the BCNF Condition	**6**	4	LL2
	b	Determine the closer of the following set of functional dependencies for a relation scheme. R(A,B,C,D,E) into R1(A,B,C) and R2(A,D,E) determine that this decomposition is a lossless-join decomposition or dependency preserving if the following set F of FD's holds: A->BC, CD->E, B->D, E->A.	**6**	4	LL3

UNIT-5

Short answer questions

Q. No.	Question	M	CO	LL
1	Define a Transaction in DBMS and list its operations.	**2**	5	LL1
2	Define DBMS Buffer and list its advantages.	**2**	5	LL1
3	List the ACID properties of a transaction.	**2**	5	LL1
4	Define System Log in DBMS and list the types of failures.	**2**	5	LL1
5	Define Schedule in DBMS and list the types of schedules.	**2**	5	LL1
6	Define File Organization and types of file organization.	**2**	5	LL1
7	Define Indexing in DBMS and State its purpose.	**2**	5	LL1
8	Differentiate between sparse index and dense index.	**2**	5	LL1
9	Differentiate between Primary index, Secondary Index	**2**	5	LL1
10	Differentiate between Indexing and Hashing.	**2**	5	LL1

Essay questions

Q. No.		Question	M	CO	LL
1	**a**	Define a Transaction. Explain in detail the operations of a Transaction with suitable examples.	**6**	5	LL2
	b	Explain Database Buffers in DBMS and its advantages.	**6**	5	LL2
2	**a**	Explain different types of failures in transaction processing.	**6**	5	LL2

	b	Explain Log-based Recovery in DBMS	**6**	5	LL2
3	**a**	Explain the various states of a Transaction with a neat diagram	**6**	5	LL2
	b	Explain the ACID Properties of Transaction with examples	**6**	5	LL2
4	**a**	Define Schedule. Explain different types of Schedules with examples.	**6**	5	LL2
	b	Consider the schedule S={R_1(A), W_2(B), W_1(A), C_1, W_2(A), C_2}. Analyze whether this schedule is Conflict Serializable. If not, list the conflicting operations that prevent conflict serializability by drawing the precedence graph.	**6**	5	LL2
5	**a**	Describe about File Organization in DBMS	**6**	5	LL2
	b	Define Indexing. Explain various Indexing Methods in DBMS	**6**	5	LL2
6	**a**	Explain Hash Based Indexing	**6**	5	LL2
	b	Explain Tree Based Indexing	**6**	5	LL2

Course Coordinator **Module Coordinator** **Programme Coordinator**

Coordinator IQAC